Diagnostics of Polarization-Squeezed Entanglement in Optical Fiber for use in Stable Quantum Sagnac Interferometry

JTM Campbell - Maynooth Dept of Electronic Engineering, Maynooth University, Republic of Ireland

***Abstract*—Squeezed light is a valuable resource in photonic quantum technology that allows us to surpass the standard quantum limit in certain measurements. This paper discusses the generation of polarization-squeezed entanglement, focusing on the generation of two-mode squeezed states entanglement from a non-linear crystal in the process of squeezing. We also examine the utilization of the GKP code, which encodes quantum information in continuous-variable systems, to provide a framework for using polarization-squeezing in manipulating quantum information in infinite-dimensional spaces. While the GKP code primarily focuses on continuous variables like position and momentum, we show that the principles of encoding and stabilizing quantum information can be extended to other types of quantum states, including polarization-encoded states. The paper aims to explore the general use of polarization squeezing as stabilizing qubits for quantum applications, such as using flying qubits for entanglement resources. We demonstrate the use of Kuramoto oscillators to simulate the use of Phase-insensitive amplification (PIA) on GKP-stabilized qubit nodes to demonstrate a technique for boosting the sensitivity of qubits-as-sensors, for instance, in an interferometer array. The ability to amplify signals between nodes without introducing noise makes PIA an essential tool for advancing our ability to detect subtle changes in stabilized qubits and unlock the potential of their use in quantum sensing technology in diverse sensor scenarios ranging from chiral polarimetry to distributed fiber-based sensing and interferometery using polarization entanglement sources. Of key import here is the means to develop new figures of merit for fiber sensing. To do this a factor, $\wr_P$, is used in relation to polarization shifts treated as a Doppler-like Effect in the network Lagrangian framework.**

I. Introduction

A. Squeezed Light and Optical Kerr Effect

Squeezed light results from reducing the uncertainty in one observable while increasing the uncertainty in its conjugate observable, as dictated by the Heisenberg uncertainty principle [1]. Nonlinear interactions such as the optical Kerr effect can be utilized to produce squeezed light in free space or in optical fibers [2].

The optical Kerr effect describes the change in the refractive index of a material in response to light intensity, I. In the context of polarization squeezing, the focus is on the nonlinearity in optical crystals such as BBO and ppKTP in nodal cavities in a network but may also focus on Kerr nonlinearities in active optical fibers, both of which enable the creation of polarization-squeezed states in a network [3].

B. Polarization Squeezing

Polarization squeezing involves manipulating the polarization states of light to achieve squeezing, particularly through the generation of two-mode squeezed states [4]. By splitting light into two orthogonal polarization axes in an optical fiber and combining the resulting squeezed beams, polarization squeezing can be achieved [5].

Consider an optical fiber with two orthogonal polarization axes. When light propagates through this fiber, it splits into two identical squeezed beams along these polarization axes. Combining these beams in a Stokes measurement results in polarization squeezing. The degree of squeezing, measured in decibels, can reach significant levels without the need for frequent adjustments [6].

Traditionally, polarization squeezing setups have been sensitive to external fluctuations and required careful tuning. However, recent developments have led to a new all-fiber setup that is simple, robust, and stable. This setup consists of passive elements only, making it easy to set up and maintain. Researchers have achieved long-term stability with more than 5 dB of squeezing, making it applicable to various quantum applications [7].

C. The GKP Code

The Gottesman-Kitaev-Preskill (GKP) code is a quantum error-correcting code that encodes quantum information into an infinite-dimensional space using continuous-variable systems [8]. In this code, a qubit is encoded in a two-dimensional subspace of an infinite-dimensional oscillator space by leveraging the properties of continuous observables and their corresponding operators.

The basic idea behind the GKP code is rooted in the properties of conjugate continuous observables, such as position q and momentum p, which satisfy the Heisenberg uncertainty principle preventing their exact simultaneous measurement [9]. However, the GKP code overcomes this limitation by introducing two operators, $\hat{S}_q$ and $\hat{S}_p$, which are periodic functions of q and p. These operators commute with each other, allowing for their joint measurement and the simultaneous encoding of quantum information.

The GKP states offer advantages in fault-tolerant quantum computing and quantum error correction by enabling the manipulation of quantum information in continuous-variable systems [10]. Their use of conjugate observables and periodic operators provides a novel approach to encoding quantum

information in infinite-dimensional spaces, paving the way for advancements in quantum information processing.

II. The GKP Code

The operators are defined as follows:

$$\hat{S}_q = e^{i2\sqrt{\pi}\hat{q}} \tag{1}$$

$$\hat{S}_p = e^{-i2\sqrt{\pi}\hat{p}} \tag{2}$$

A. Encoding Quantum Information

By stabilizing the qubit subspace using the operators, the GKP code encodes quantum information in a structured manner within an infinite-dimensional oscillator space.

To examine the dynamics of polarization-entangled states using GKP-encoded qubits, we can represent the total Hamiltonian of the system as:

$$\hat{H} = \hat{H}_0 + \hat{H}_{int}, \tag{3}$$

where $\hat{H}_0$ is the Hamiltonian governing the individual oscillators and $\hat{H}_{int}$ represents the interaction terms that couple the qubits.

The time evolution of these states can be described using the Schrödinger equation:

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle. \tag{4}$$

Using the phase operators $\hat{S}_q$ and $\hat{S}_p$, we can analyze how the GKP states evolve over time under phase rotations and examine the resulting characteristics of the entangled qubits.

The GKP states can be visualized as an infinite, periodic lattice structure in the phase space, characterized by the spacing $\sqrt{\pi}\sigma$ in position space, resulting in a periodicity in both q and p. If we denote the lattice points by (q_n, p_m), we find that:

$$q_n = n\sqrt{\pi}\sigma, \tag{5}$$

$$p_m = m\frac{\sqrt{\pi}\hbar}{\sigma}, \tag{6}$$

(where n, m are positive and real).

The GKP states are robust against certain types of errors due in large part to their periodic nature. The logical operators that act on the GKP states can be represented in the phase space as translations:

$$\hat{T}_q(a) = e^{-\frac{ia}{\hbar}\hat{p}}, \tag{7}$$

$$\hat{T}_p(b) = e^{-\frac{ib}{\hbar}\hat{q}}, \tag{8}$$

which further interact with the lattice structure by adding or shifting these grid points.

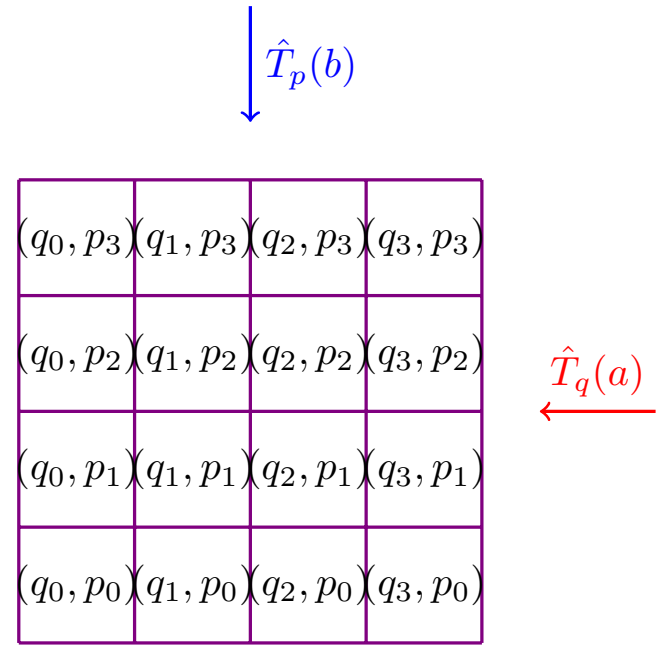


Fig. 1. An associated labeled grid network diagram using the GKP states: The GKP states can be visualized as an infinite, periodic lattice structure characterized by the spacing $\sqrt{\pi}\sigma$ in position space, resulting in a periodicity in both q and p. If we denote the lattice points by (q_n, p_m), we find that $q_n = n\sqrt{\pi}\sigma$, $p_m = m\sqrt{\frac{\pi\hbar}{\sigma}}$, (where n, m are positive and real).

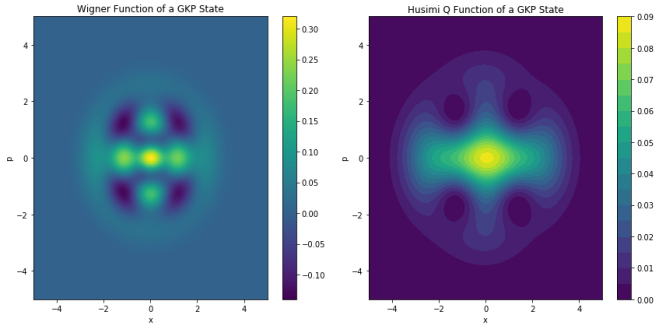


Fig. 2. Wigner and Husimi phase space functions of GKP states. Structured representation of quantum information in the phase space offers promising applications in fault-tolerant quantum computation and quantum error correction.

III. Quantum GKP Encoding

Similarly to quadrature amplitude modulation (QAM) encoding used in classical communication, we may use the quantum GKP encoding to correct loss errors. The basic idea behind the GKP code [8] is that while the exact value of two conjugate continuous observables cannot be measured simultaneously, the two operators

$$\hat{q} = \text{quadrature operator} \tag{9}$$

and

$$\hat{p} = \text{momentum operator} \tag{10}$$

which are periodic functions of $\hat{q}$ and $\hat{p}$, commute with each other and therefore can be measured simultaneously. The

GKP code thus encodes a qubit in a two-dimensional subspace of an infinite-dimensional oscillator space. This subspace is stabilised by these two operators, and the GKP state can be visualised as an infinite, periodic grid structure in the (q, p) phase space. For the GKP code based on a square lattice, which we will consider here and in our simulation (describing the square lattice of oscillators as a Kuramoto Quantum Oscillator system), the standard basis states are given as:

$$|0\rangle = \sum_{n=-\infty}^{\infty} |q = 2n\sqrt{\pi l}\rangle \tag{11}$$

Similarly, the GKP X basis logical states are:

$$|+\rangle = \sum_{n=-\infty}^{\infty} e^{i\pi np/2} |q = 2n\sqrt{\pi l}\rangle \tag{12}$$

We can see that the grid corresponding to the basis state $|0\rangle$ is shifted by Δq along the $\hat{p}$ quadrature with respect to $\hat{q}$. Hence, by measuring the two stabilisers, which amounts to measuring both $\hat{q}$ and $\hat{p}$ quadratures of the GKP state modulo $\sqrt{\pi l}$, we can detect and correct any small shifts (of size smaller than Δq) in both quadratures, in a way that does not reveal the encoded logical information.

The two GKP stabilisers can, in fact, be measured using additional GKP ancilla modes through a Steane error-correction process. Application of a two-mode operation between the GKP data mode and the GKP ancilla can transfer the information about the noise from the data qubit onto the ancilla in such a way that the logical information is not revealed. Hence, measuring the ancilla and applying a feedback displacement based on the measurement outcome enables GKP quantum error correction.

IV. Phase-Insensitive Amplification (PIA) on GKP Qubits

In the considered architecture, quantum information encoded in the GKP qubits is sent through the repeater chain as follows. After Alice performs the encoding operation, she applies the phase-insensitive amplification and sends the GKP qubit through the lossy channel towards the first GKP repeater. The repeater performs GKP correction first in $\hat{q}$ and then in $\hat{p}$ quadrature. After that, it again applies the phase-insensitive amplification and sends the state to the next repeater. In this way, the encoded qubit can effectively be transmitted to Bob. In our model, we consider two sources of imperfections apart from loss in the communication channel. Firstly, we assume a finite photon in-coupling and out-coupling efficiency η_0, which quantifies the efficiency of transferring the photon from the fiber to the repeater and back into the fiber. Hence, the total transmissivity of the lossy channel between two neighbouring repeaters separated by the distance L is:

$$T(L) = \exp\left(-\frac{L}{L_0}\right) \tag{13}$$

where L_0 is the attenuation length of the channel. Here, we assume transmission at telecom frequency at which $L_0 = 22$ km. The second imperfection we consider is the finite amount of GKP squeezing. Under finite squeezing, the GKP grid does not consist of delta functions but of Gaussian peaks with an overlying envelope function, such that the peaks in the Wigner function decay to zero height in the limit of large quadrature values. The standard deviation of these finitely squeezed Gaussian peaks is given by σ_{GKP}, and the amount of squeezing can also be quantified by comparing σ_{GKP} to the standard deviation of a Gaussian peak of a coherent state given by σ_{coh}. Hence, squeezing expressed in dB can be defined as:

$$\text{Squeezing (dB)} = 10\log_{10}\left(\frac{\sigma_{\text{coh}}}{\sigma_{\text{GKP}}}\right) \tag{14}$$

In our analysis, similarly to [8], we consider a conservative error model that allows us to describe a finitely squeezed GKP qubit as an ideal GKP state subjected to a random displacement according to a probability distribution parameterised by σ_{GKP}. Since both finite GKP squeezing and the channel noise lead now to random displacement errors, we can reliably approximate the repeater performance by considering perfect error correction using infinitely squeezed ancillas, which, however, is now performed on the data qubits subjected to an effective communication channel. This effective channel now includes not only the noise coming from η defined in Eq. (4) but also from non-zero σ_{GKP}. That is, we consider an approximation in which the noise from finite squeezing can be effectively incorporated into the channel and combined with the noise due to photon loss. This approximation enables us to construct a simple analytical model through which we can efficiently evaluate repeater performance, including optimisation over repeater spacing. We validate this analytical model against a numerical Monte Carlo simulation.

V. Self-Stabilizing Qubits

The concept of self-stabilizing qubits involves using polarization-squeezed entanglement as resources for stabilizing quantum information encoded in continuous variables. Here, we discuss the implementation of the Gottesman-Kitaev-Preskill (GKP) quantum error correction code for error correction in continuous-variable quantum systems.

Specifically, time-domain multiplexing involves dividing continuous light into discrete time segments, encoding quantum information in separated pulses, thus enabling scalable qubit numbers along the temporal axis. Furthermore, generating large-scale entangled states—such as two-dimensional cluster states—can be achieved with just a handful of squeezed light sources, optical delay lines, and beam splitters. This approach circumvents the need for extensive integration or large-scale parallelization, offering a practical pathway toward universal quantum computation with manageable equipment.

Quantum error correction schemes based on photon parity and continuous variables, by Valentina Parigi et al in research on CV quantum networks at telecommunication wavelengths has also revealed that even a single nonlinear crystal can create many-body bosonic modes that can be treated as a networked cluster state.

Using this one can also be borrow stabilization techniques from the quantum computing literature to further enhance

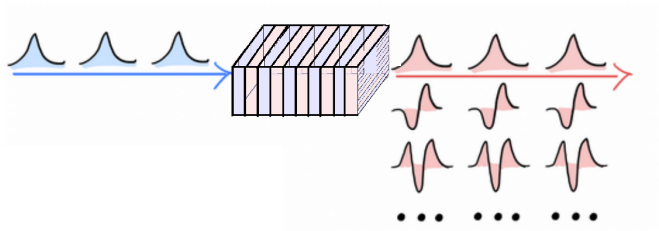


Fig. 3. Schematic of a CV entanglement source as used by Parigi et al in developing networked cluster states using bosonic modes

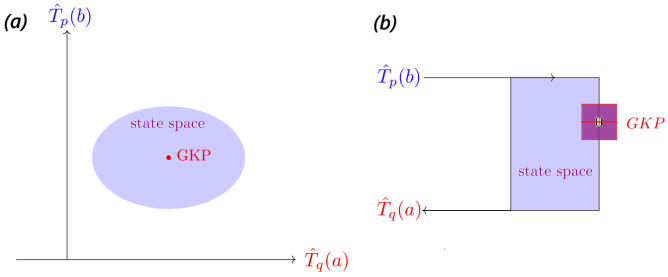


Fig. 4. An auxiliary GKP qubit (red) introduced into a quantum error correction or encoding scheme can assist in maintaining the stabilizer structure while expanding the accessible state space (purple) in fig 4(a). Unlike the primary logical qubits that encode the information in quantum computing, the aid qubit is not directly measured during the error detection process in fig 4(b). Instead, it serves as a resource to facilitate more robust interactions and operations, helping to preserve the error-detecting properties of the stabilizer code.

the robustness of fiber optical quantum networks. This brings about some familiar stabilization techniques, involving surface codes, along with new interpretations involving synchronization used in classical networks.

The GKP qubit here can be an ancilla or "aid" qubit in stabilizing network structure as is used in some quantum error correction schemes in computation involving GKP encoding.

VI. GKP Quantum Error Correction Methods applied to Networking or Fault-Tolerence

Error correction in GKP qubits involves interaction between data and ancilla modes, followed by measurement to detect errors without revealing encoded information.

an auxiliary GKP qubit introduced into a quantum error correction or encoding scheme can assist in maintaining the stabilizer structure while expanding the accessible state space. Unlike the primary logical qubits that encode the information in quantum computing, the aid qubit is not directly measured during the error detection process. Instead, it serves as a resource to facilitate more robust interactions and operations, helping to preserve the error-detecting properties of the stabilizer code.

This design allows the system to detect and potentially correct errors more effectively by expanding the overall state space, providing additional degrees of freedom for encoding and error management. The aid qubit's unmeasured status ensures that it does not collapse the quantum state prematurely, thereby maintaining coherence and enabling more resilient quantum interaction encoding schemes.

This network illustrates that an error (shift) occurs on the data mode, which interacts with an ancilla mode via a two-mode operation. Measuring the ancilla provides syndrome information for error detection and correction, all while preserving the encoded quantum information.

A. GKP Encoding

The GKP encoding scheme encodes quantum information in a two-dimensional subspace of an infinite-dimensional oscillator space by exploiting the commutation relations of two conjugate continuous observables. The GKP code stabilizes the encoded qubit against small displacements by measuring two stabilizer observables. The GKP basis states are given as follows:

$$|0\rangle = \sum_{n=-\infty}^{\infty} |q = 2n\sqrt{\pi l}\rangle \tag{15}$$

The GKP X basis logical states are:

$$|+\rangle = \sum_{n=-\infty}^{\infty} e^{i\pi np/2} |q = 2n\sqrt{\pi l}\rangle \tag{16}$$

B. GKP Quantum Error Correction

Error correction in GKP qubits can be achieved through the measurement of GKP stabilizers using ancilla modes. Applying a two-mode operation between the data mode and the ancilla allows for error detection and correction without revealing the encoded information.

VII. Simulation of GKP in Phase Space using a Grid of Kuramoto Oscillators

The GKP-encoded qubits from two Entanglement Resources, Rho and Sigma, are sent through a series of PM fibers, a combiner, and a splitter by performing GKP correction and phase-insensitive amplification at each repeater station to combat losses in the communication channel. Imperfections such as finite photon in-coupling efficiency and finite GKP squeezing are considered in the model. The transmissivity of the lossy channel between repeaters separated by distance L is given by:

$$T(L) = \exp\left(-\frac{L}{L_0}\right) \tag{17}$$

where L_0 is the attenuation length of the channel. The amount of squeezing in dB is calculated as:

$$\text{Relative Squeezing (dB)} = 10\log_{10}\left(\frac{\rho_{\text{GKP}}}{\sqrt{2}}\right) \tag{18}$$

In our scenario, we associate the phase of each Kuramoto oscillator with the phase of the GKP states that are transmitted through the PM fiber. The model can be described as a network graph of oscillators using the following equations.

Network Graph of Quantum Harmonic Oscillators as a Reduced Density Matrix

Overall, it is possible to represent a network with given adjacency and network Laplacian matrices as a reduced cavity density matrix. From this, it is possible to compute an operational meaning of the trace distance between the EPR pair as the source density matrix and the network to which the density matrix is operationally coupled. First, let us create a reduced cavity density matrix from a network Laplacian. Given a network graph G:

$$M(G)_{(i,j),k} = \begin{cases} 1 & \text{if } v_k = v_i \\ -1 & \text{if } v_k = v_j \\ 0 & \text{otherwise} \end{cases} \tag{19}$$

We get the following formula to retrieve a density matrix given a graph Laplacian L:

$$\rho(G) := \frac{L(G)}{\mathrm{Tr}[L(G)]} \tag{20}$$

which can be expressed in terms of Z as:

$$\widetilde{\rho}(G) := \frac{\mathrm{e}^{-\tau L(G)}}{Z} \tag{21}$$

where

$$Z = \mathrm{Tr}\left[\mathrm{e}^{-\tau L(G)}\right] \tag{22}$$

Oscillator Node Dynamics

Each Kuramoto oscillator (representing a GKP state) evolves according to the following equation:

$$\frac{d\phi_i}{dt} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\phi_j - \phi_i) - \gamma\phi_i \tag{23}$$

where:

- ϕ_i is the phase of the i-th oscillator,
- ω_i is the natural frequency of the i-th oscillator,
- K is the coupling strength between the oscillators,
- N is the total number of oscillators,
- γ represents a damping factor accounting for channel losses.

Transmissivity Incorporation

Incorporate the effect of transmissivity $T(L)$ into the oscillator dynamics, reflecting how these oscillators' coupling is affected by the distance L:

$$K(L) = K_0 \cdot T(L) = K_0 \cdot \exp\left(-\frac{L}{L_0}\right) \tag{24}$$

where K_0 is the original coupling strength without distance attenuation.

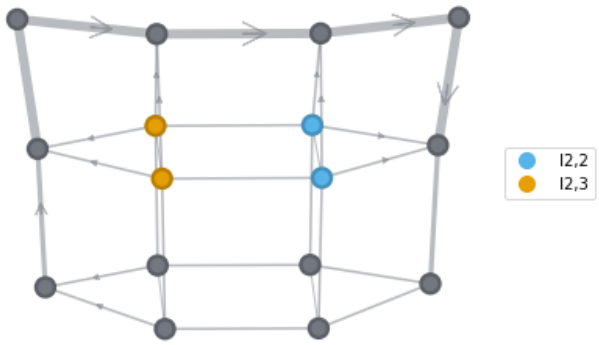


Fig. 5. The contracted tensor network with scalar values in directional acyclic connections to the bra and ket PEPs (projected enagled pair states)

A. Inclusion of Squeezing

The effect of relative squeezing can alter the natural frequencies of the oscillators. A squeezed oscillator can be modeled as having its natural frequency adjusted based on the relative squeezing:

$$\omega_i' = \omega_i + R_i \tag{25}$$

with the squeezing term given by the formula:

$$R_i = -\frac{\kappa}{2} \cdot \mathrm{RelativeSqueezing}(dB) \tag{26}$$

where κ is a scaling factor reflecting how squeezing influences the frequency of the oscillators.

Hyper-Dimensional Simulation

If we have multiple distances (repeaters) and encodings, we can model this in a higher-dimensional phase space: These logical operators interact with the lattice architecture of GKP states, facilitating the addition or shifting of grid points within the lattice framework. Consequently, the action of these operators can be conceptualized in terms of oscillatory behaviors, analogous to the dynamics observed in higher-order networks of Kuramoto oscillators.

In the context of GKP surface codes, we define a higher-dimensional phase space that accommodates multiple distances and encoding strategies. The dynamics of this higher-order Kuramoto system can be expressed as follows:

$$\frac{d\phi_{i,j}}{dt} = \omega_{i,j}' + \frac{K(L_j)}{N}\sum_{k=1}^{N}\sin(\phi_{k,j} - \phi_{i,j}) - \gamma\phi_{i,j} \tag{27}$$

Here, $\phi_{i,j}$ represents the phase of the i-th oscillator situated at the j-th repeater, where each oscillator is tied to the intricate structure of the GKP lattice.

This model was run on a contracted tensor network on QuIMB in python.

Importantly, this higher-order Kuramoto model not only captures the synchronization dynamics among oscillators affiliated with nodes but also extends to the synchronization of

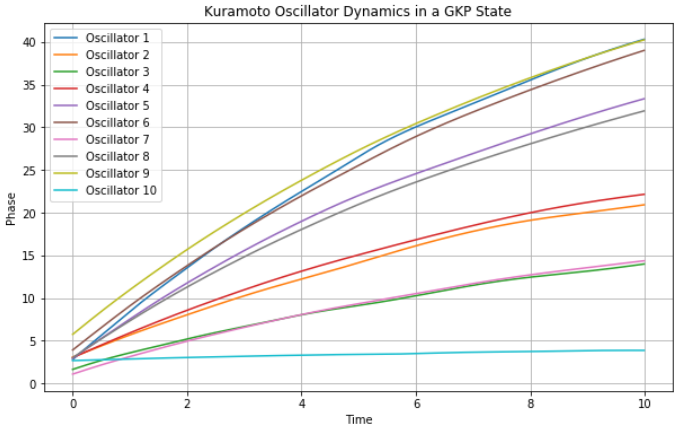


Fig. 6. Kuramoto oscillator dynamics in a GKP state

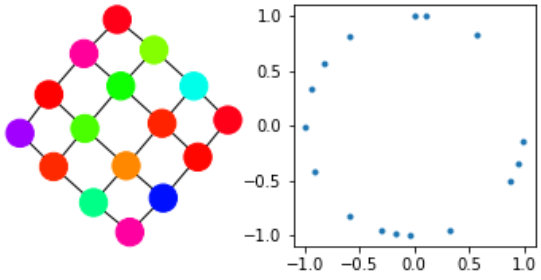


Fig. 7. (Left to Right) GKP as a network graph and Kuramoto oscillator dynamics in phase space

topological signals within simplicial complexes. In this vein, oscillators represent not just nodes but also higher-dimensional simplices—such as links and triangles—that characterize a type of surface code in a GKP configuration.

This framework unveils an interplay within topological signals of varying dimensions, revealing their capacity to coexist and engage in non-trivial coupling. Simulating Kuramoto dynamics in a GKP state in this manner has allowed for the study of how these entangled pairs in a GKP lattice may influence network synchronization, providing a quantum-level understanding of phase locking and collective behavior, providing motivation to develop such GKP stabilization as a fiber optic setup.

Such dynamics can lead to simultaneous explosive transitions, akin to the percolation thresholds observed in classical networks. These phenomena underscore the intricate connections between complex systems, where topology governs the emergent behavior, showcasing how GKP states and their logical operators can serve as powerful models for exploring synchronization and phase transitions in multifaceted systems.

B. Experimental Transmission through a PM Fiber System

The use of polarization squeezing for self-stabilizing qubits in continuous-variable quantum systems shows promise for quantum error correction and communication applications. By incorporating finite squeezing and channel imperfections into the analysis, we can optimize tomography performance in an operationally realistic scenario.

In the context of polarization squeezing, the focus is on the nonlinearity in optical crystals such as BBO and ppKTP in nodal cavities in a network, but may also focus on Kerr nonlinearities in active optical fibers. Both of these enable the creation of polarization-squeezed states in a network.

Consider an optical fiber with two orthogonal polarization axes. When light propagates through this fiber, it splits into two identical squeezed beams along these polarization axes. Combining these beams in a Stokes measurement results in polarization squeezing. The degree of squeezing achieved can be quantified in decibels (dB). Researchers have reported more than 5 dB of directly measured squeezing using robust setups without the need for frequent adjustments.

This work investigates how phase synchronization among fiber network nodes, quantified by the parameter R, the degree of synchronization between nodes in a network from the Kuramoto model and how R between Sagnac interferometer nodes influences the polarization properties of transmitted light between these interferometers. Using concepts from oscillator synchronization, coherence, and polarization of optical fields, we develop a physical intuition connecting the degree of synchronization R , the State of Polarization (SoP), and the Degree of Polarization (DoP).

In fiber networks containing Sagnac interferometers as stabilizer nodes, phase coherence between these nodes leads to transfer of stable polarization states, essential for some forms of information transfer being explored for classical and quantum networking. When

$$R \to 1$$

, the fields are phase-aligned, producing high DoP, whereas

$$R \to 0$$

results in incoherent superpositions with decreased polarization purity.

Understanding this connection offers a pathway to optimize polarization entanglement distribution and polarization stabilization in quantum networks.

In the Kuramoto Model for N oscillator nodes in a network the degree of synchronization between nodes is given by the general formula:

$$Re^{i\psi} = \frac{1}{N}\sum_{k=1}^{N} e^{i\theta_k} \tag{28}$$

where:

- θ_k: phase of the k^{th} oscillator
- $R \in [0, 1]$: magnitude indicating synchronization.

Considering polarized light is an Electric Field Superposition

$$E(t) = \sum_{k=1}^{N} |E_k| e^{i(\omega t + \phi_k)} \mathbf{e}_k \tag{29}$$

- $|E_k|$: amplitude
- ϕ_k: phase
- $\mathbf{e}_k$: polarization vector

High R between synchronized polarized light nodes in fiber loops implies ϕ_k are locked, leading to stable polarization; low R causes incoherence and polarization fluctuations.

- When phases are aligned ($R \to 1$), the superposed electric field is highly coherent, resulting in a well-defined polarization state with high DoP.
- When phases are random ($R \to 0$), the superposition is incoherent, leading to depolarization or unpolarized light with low DoP.
- The superposition can be seen as a weighted combination of polarization states:

$$\boxed{\mathbf{S}_{\text{super}} \propto R\,\hat{\mathbf{S}}_A + (1-R)\,\hat{\mathbf{S}}_B}$$

with R representing the degree of phase coherence.
Define fixed states:

$$\begin{aligned} \mathbf{S}_A &= [1,1,0] \quad \textbf{(Horizontal linear)} \\ \mathbf{S}_B &= [1,0,0] \quad \textbf{(Circular right)} \end{aligned} \tag{30}$$

Normalize:

$$\hat{\mathbf{S}}_A = \frac{\mathbf{S}_A}{\|\mathbf{S}_A\|}, \quad \hat{\mathbf{S}}_B = \frac{\mathbf{S}_B}{\|\mathbf{S}_B\|}$$

Choose R values:

$$\{1, 0.9, 0.7, 0.5, 0.3, 0.1, 0\}$$

Initialize list

superpositions

for each R in R-values Compute superposition
Results of the simulation here

1) Robust All-Fiber Setup: Traditionally, polarization squeezing setups have been sensitive to external fluctuations and required careful tuning. Recent developments have led to a new all-fiber setup that is simple, robust, and stable. This setup consists of passive elements only, making it easy to set up and maintain using GKP coded states.

The proposed experiment can leverage a polarization-squeezed state generated from a BBO Spontaneous Parametric Down-Conversion (SPDC) source of light, incorporated into an optical fiber Sagnac interferometer. The characteristics of the Sagnac interferometer will allow for the measurement of the polarization states and enable the exploration of the effects of polarization squeezing within the fiber system.

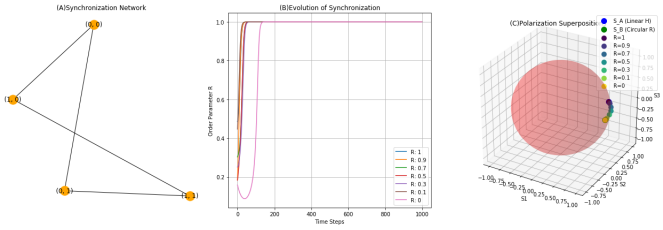


Fig. 8. (A) Simulation of a Network of Sagnac Loops in Synchronization, with each loop carrying 2 States of Polarization(B) Evolution of Synchronization over a list of degrees of syn chronization, R, in the Kuramoto Model (C) The two fixed polarization states, on the Poincaré sphere. Showing how their superposition changes as R varies from 1 (full coherence) to 0 (no coherence). Illustrating the resulting superposed Stokes vector for different R values. The blue point at the equator (linear horizontal). The green point at the north pole (right circular). As R decreases, the superposition point moves from the Sa position towards a mixture closer to Sb, eventually collapsing towards the middle (less polarized state).When R=1, the superposition coincides with Sa; when R=0, it coincides with Sb.

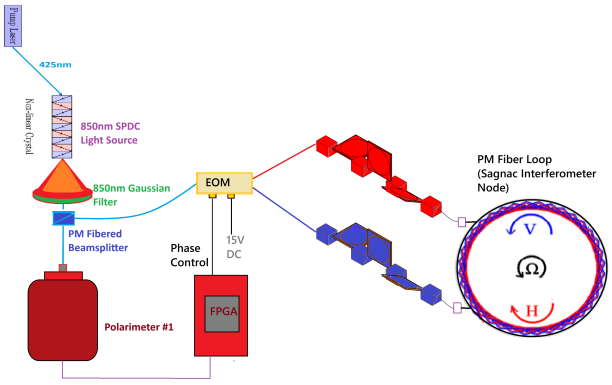


Fig. 9. All fiber setup diagram

Traditionally, polarization squeezing setups have been sensitive to external fluctuations and required careful tuning. Recent developments have led to a new all-fiber setup that is simple, robust, and stable. This setup consists of passive elements only, making it easy to set up and maintain using coded states. Active tuning is possible however to mitigate external noises, using an FPGA and EOM for phase control (and in principle synchronization between muliple Sagnac nodes in a real network.

The proposed experiment can leverage a polarization-squeezed state generated from a BBO Spontaneous Parametric Down-Conversion (SPDC) source of light, incorporated into an optical fiber Sagnac interferometer. The characteristics of the Sagnac interferometer will allow for the measurement of the polarization states and enable the exploration of the effects of polarization squeezing within the fiber. Even environmental effects, both passive such as orientation to Earth's rotation Ω , and/or seismic rotational effects.

VIII. Polarization Shift Relation to Action Derivatives

Based on the established relation between frequency shifts and the derivative of the action [13], (i.e. the Doppler effect

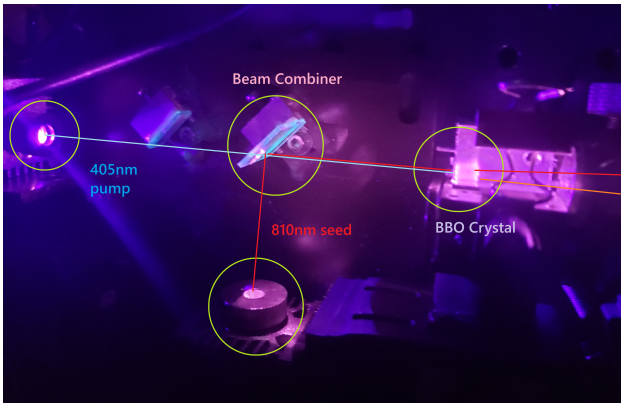


Fig. 10. Non-linear optics entangled light source. Based on pumping a BBO crystal with a 405nm pump laser and 810nm seed laser to generate 2 810nm entangled photon pairs

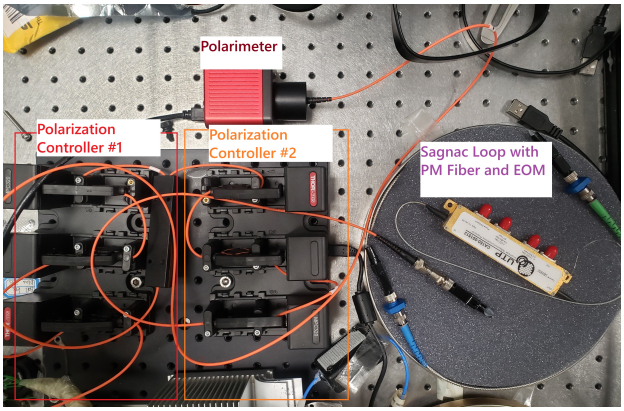


Fig. 11. Polarization controllers of signal and idler beams before coupling to an EOM and Sagnac fiber loop. Reflected beams are analysis using a stokes rotating waveplate polarimeter

applied to the network Lagrangian) we propose that the polarization shift

$$\Delta P$$

in a fiber network can be modeled as being proportional to the rate of change of the networked systems action S:

$$\boxed{\Delta P \propto \mathfrak{Q}_P \frac{\partial S}{\partial t}}$$

where

$$\mathfrak{Q}_P$$

is a proportionality constant encapsulating how sensitive the polarization state is to variations in the physical parameters affecting the system. This relation suggests that physical disturbances—such as stress, temperature fluctuations, or fiber imperfections—that alter the underlying action will directly influence the polarization state of the transmitted light.

$\mathfrak{Q}_P$ in Fiber Networks is thus an emergent, phenomenological parameter that links temporal changes in physical system action to observable polarization shifts.

It's not a textbook quantity yet — but in the context of our framework, it serves as a diagnostic proxy for how sensitive any polarization maintaining fiber system is to underlying physical disturbances.

A. Computational Extraction of $\mathfrak{Q}_P$ from PAX1000 Polarimeter Data

To experimentally validate the proposed polarization-action relationship, we developed a computational pipeline that leverages real-time measurements from a Thorlabs PAX1000 polarimeter. The device outputs time-resolved Stokes parameters either directly or indirectly via measured azimuth (ψ) and ellipticity (χ) angles, which characterize the polarization ellipse.

These angles are mapped onto the Poincaré sphere using the well-established relations:

$$S_1 = \cos(2\psi)\cos(2\chi), \quad S_2 = \sin(2\psi)\cos(2\chi), \quad S_3 = \sin(2\chi) \tag{31}$$

This reconstruction enables the visualization of the evolving polarization state as a trajectory on the Poincaré sphere. The angular difference between consecutive polarization states, $\Delta P(t)$, is computed as:

$$\Delta P(t) = \cos^{-1}\left(\mathbf{S}(t) \cdot \mathbf{S}(t - \Delta t)\right) \tag{32}$$

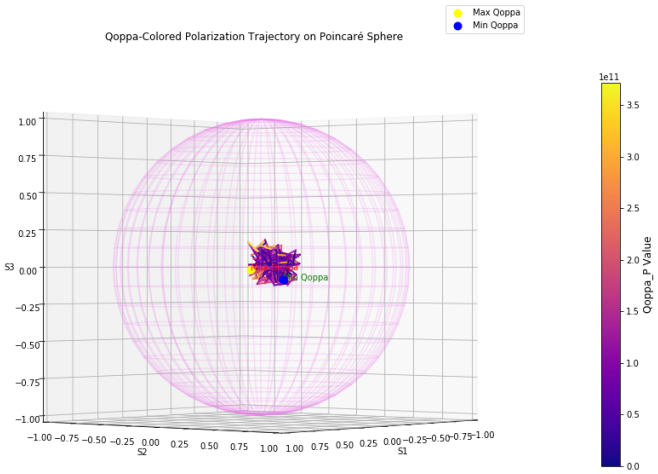
Simultaneously, a proxy for the time derivative of the action, $\partial S/\partial t$, is computed from the norm of the Stokes vector:

$$|\mathbf{S}(t)| = \sqrt{S_1^2 + S_2^2 + S_3^2}, \quad \frac{\partial S}{\partial t} \approx \frac{d}{dt}|\mathbf{S}(t)| \tag{33}$$

Combining both, the dimensionless phenomenological factor $\mathfrak{Q}_P(t)$ is estimated by:

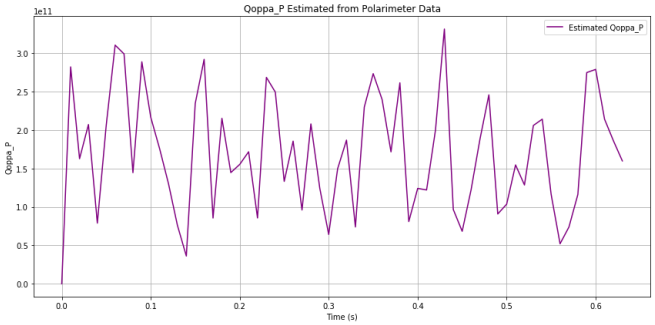
$$\mathfrak{Q}_P(t) = \frac{\Delta P(t)}{\partial S/\partial t + \varepsilon} \tag{34}$$

where ε is a small regularization constant to avoid numerical instability.

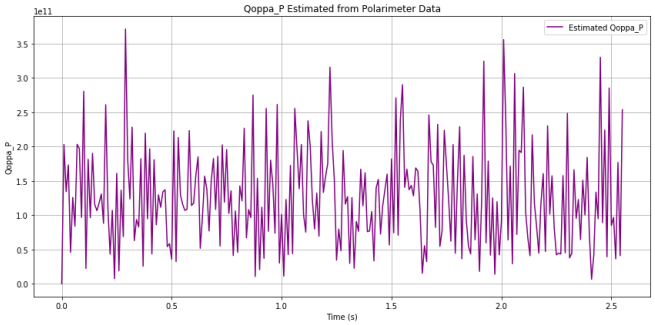


This method allows the real-time estimation of $\mathfrak{Q}_P$ solely from polarization measurements, enabling physical insight into the dynamic sensitivity of a fiber network to external perturbations. Furthermore, by mapping $\mathfrak{Q}_P(t)$ values onto the Poincaré sphere as a color-indexed trajectory, one can visually correlate high-sensitivity events with specific polarization regions.

Moreover by monitoring $\mathfrak{Q}_P$ over time much more simpler time-series information can be extracted instead of more complex and costly monitoring of polarization state trajectories on the Poincaré sphere. This can be seen with 64-bit samplings in an experimental quantum key exchange scenarios.



and 256-bit samplings in a similar experimental scenario



For real world QKD, this would be a useful diagnosis to make, as traditional polarization state monitoring and sampling is computationally costly using any polarimeter, often involving interruptions to the key exchange or additional multiplexed alignment channels, which ultimately reduce key rate and increase system complexity [16].

For applications such as network reconstruction, reducing system observations to a simple scalar time series greatly simplifies the process—Takens' embedding theorem then allows the reconstruction of a dynamical phase-space, providing a foundation for building recurrence networks and quantum network-based mappings [17].

Physical Consequences

In fiber networks, this relation highlights the importance of controlling environmental and physical factors, as they can induce changes in the system's action, leading to polarization drift or scrambling. Such polarization shifts can degrade the performance of polarization-sensitive applications, including high-speed communications, quantum key distribution, and coherent detection schemes. Understanding and modeling the relation between action derivatives and polarization can help develop more robust fiber systems with enhanced stability against external perturbations.

B. Diagnostic Role of $\mathfrak{Q}_P$ in Fiber Networks

The phenomenological coefficient $\mathfrak{Q}_P$, introduced via the relation $\Delta P(t) \propto \mathfrak{Q}_P \, \partial S/\partial t$, plays a crucial role in diagnosing physical disturbances in fiber-optic systems. It encapsulates how sensitive a given polarization trajectory is to changes in the field-based action $S(t)$, which itself depends on environmental and structural perturbations. This subsection outlines key applications of $\mathfrak{Q}_P$ and compares its advantages with standard polarization metrics.

TABLE I
APPLICATIONS OF $\mathfrak{Q}_P$ IN FIBER NETWORK DIAGNOSTICS

Application Area	**Role of $\mathfrak{Q}_P$**	**Example Use Case**
Fiber Stress Detection	Tracks microbending or compression via local birefringence changes	Detecting mechanical strain or tight bends in racks
Thermal Gradient Sensing	Captures refractive index fluctuations via time-varying permittivity	Identifying hot spots in buried or aerial fiber
Vibration and Acoustic Monitoring	Detects mode coupling via polarization shift feedback	Intrusion detection through vibration response
Birefringence Mapping	Indicates spatial birefringence variation affecting PMD	Identifying high-PMD regions in deployed fiber
Coherent System Diagnostics	Links drift in phase/polarization to external perturbations	Diagnosing polarization instability in QKD systems
Aging and Degradation Detection	Monitors long-term sensitivity drift over time	Fiber health tracking in legacy or submarine systems

TABLE II
COMPARISON OF $\mathfrak{Q}_P$ WITH TRADITIONAL POLARIZATION METRICS

Metric	**Limitation**	**What $\mathfrak{Q}_P$ Adds**
SOP Drift	Reports change but not cause	Infers sensitivity to action variations
DOP (Degree of Polarization)	Cannot detect variation in fully polarized light	Captures real-time dynamics
PMD (Polarization Mode Dispersion)	Static or low-frequency analysis	Enables real-time local sensing
BER/SNR	Reports impact, not mechanism	Links underlying physical changes to polarization effects

This formulation suggests that $\mathfrak{Q}_P$ could serve as a foundational parameter for adaptive control and monitoring in both classical and quantum fiber-optic systems. By monitoring $\mathfrak{Q}_P(t)$, one can infer not just that polarization has shifted, but that a physical perturbation of interest has occurred—enabling predictive diagnostics, failure anticipation, and closed-loop stabilization strategies.

IX. SOP AND DOP MEASUREMENTS USING A POLARIMETER AND FPGA WITH EOM PHASE QUBIT TUNING

For tuning the EOM, a Red Pitaya StemLab-10 was used. This platform also allowed for Self-homodyne detection, eliminating the requirement of a local oscillator, wherin the entangled optical signal mixes with a delayed version of itself in detection.

The red pitaya oscilloscope ap also allows us to view the phase space trajectory of the self-homodyne detector while we can monitor the polarization using a thorlabs polarimeter.

From our point of view, the upper bound of state of polarization vs degree of polarization measures (SoP vs DoP) can serve as a measure of the error level of a quantum channel. a higher upper bound ensures a lower probability of information leakage due to error or de-synchronization of the measurement setup, i.e. by problems with Sagnac node synchronization.

The phase tuning of the EOM can be measured by Homodyne detection of the state using a dual polarimetry modification of the setup.

A. Interpretation of Qoppa Bounds in the Context of Channel Stability

The Qoppa parameter, $\mathfrak{Q}_P$, derived from polarization measurements, serves as a dynamic indicator of channel behavior in a quantum communication system. By computing upper and lower bounds on $\mathfrak{Q}_P$ over time, we can capture fluctuations that may correspond to either entangled state preservation or loss due to decoherence or dephasing.

The **lower bound** of $\mathfrak{Q}_P$ corresponds to states with minimal change in polarization, implying stable and synchronized evolution of the system's polarization parameters. In this regime, the quantum channel can be considered robust, with entanglement exchange largely preserved.

Conversely, the **upper bound** indicates rapid, possibly erratic changes in the polarization state, suggesting potential de-synchronization or depolarization of the quantum channel. When $\mathfrak{Q}_P$ trends persistently toward its upper bound, it signals a breakdown in entanglement transfer, typically associated with environmental noise, drift, or lack of synchronization between communicating parties. On the Poincaré sphere, this manifests as trajectories that span larger angular displacements or evolve with increasing randomness.

In this context, the upper and lower bounds of $\mathfrak{Q}_P$ offer a visual and analytical mechanism for diagnosing the operational health of polarization-based quantum links.

The lower bound can provide a measure of the potential failure of a quantum communication system, wherein de-synchronization has effectively collapsed the exchange of entanglement, i.e. by depolarization of the channel, which can be seen as a trend towards the maximum value of Qoppa mapped to the polarization trajectories on the Poincare sphere.

B. Interpretation of Qoppa Bounds in the Context of Channel Stability

The Qoppa parameter, $\mathfrak{Q}_P$, derived from polarization measurements, serves as a dynamic indicator of channel behavior in a quantum communication system. By computing upper and lower bounds on $\mathfrak{Q}_P$ over time, we can capture fluctuations that may correspond to either entangled state preservation or loss due to decoherence or dephasing.

The **lower bound** of $\mathfrak{Q}_P$ corresponds to states with minimal change in polarization, implying stable and synchronized evolution of the system's polarization parameters. In this regime, the quantum channel can be considered robust, with entanglement exchange largely preserved.

Conversely, the **upper bound** indicates rapid, possibly erratic changes in the polarization state, suggesting potential de-synchronization or depolarization of the quantum channel. When $\mathfrak{Q}_P$ trends persistently toward its upper bound, it signals a breakdown in entanglement transfer, typically associated with environmental noise, drift, or lack of synchronization between communicating parties. On the Poincaré sphere, this manifests as trajectories that span larger angular displacements or evolve with increasing randomness.

The following figures illustrate this relationship for two different data sets, corresponding to 64-bit key exchanges and 256-bit key exchanges in the experimental quantum key exchange scenarios.

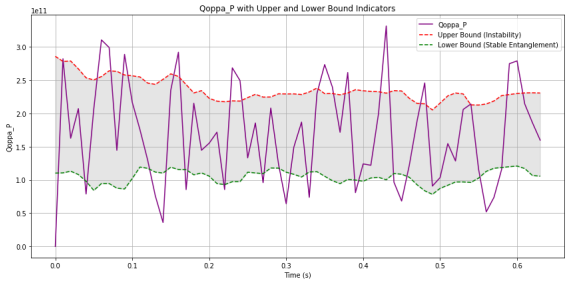


Fig. 12. Time-resolved Qoppa parameter $\mathfrak{Q}_P$ for dataset 64, showing upper and lower bounds as indicators of channel stability. The shaded region highlights fluctuations within the bound envelope.

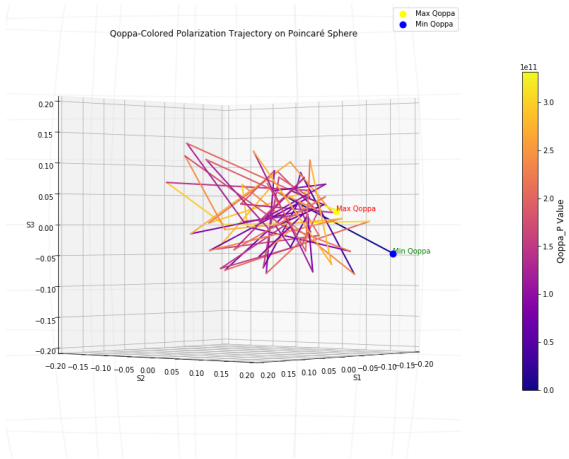


Fig. 13. Corresponding polarization trajectory on the Poincaré sphere for dataset 64, colored by $\mathfrak{Q}_P$ values. Trajectories near the sphere's surface with smooth transitions indicate low $\mathfrak{Q}_P$ (stable), while larger or more erratic movements correspond to higher values (potential de-synchronization).

As shown in Figures 12 and 14, the Qoppa bounds act as indicators of the system's dynamical regime. When the $\mathfrak{Q}_P$ trajectory remains near its lower bound, the system appears stable. In contrast, approach to the upper bound reflects increasing instability, as further emphasized in the Poincaré mappings (Figures 13 and 15), which visualize polarization trajectory complexity and drift in relation to Qoppa.

Fig. 14. Time-resolved Qoppa parameter Υ_P for dataset 256, again showing bounded behavior. The approach to the upper bound reflects increased polarization instability.

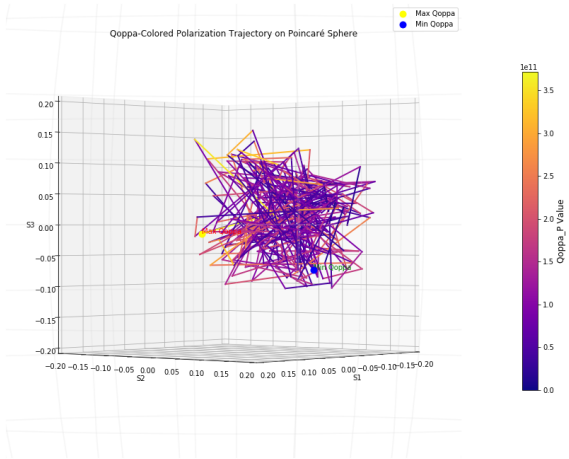


Fig. 15. Poincaré sphere trajectory for dataset 256, colored by Υ_P. The presence of large angular displacements and more disordered motion corresponds to high Υ_P values, signaling degraded entanglement exchange.

In this context, the upper and lower bounds of Υ_P offer a visual and analytical mechanism for diagnosing the operational health of polarization-based quantum links.

C. Interpretation of Qoppa in the Bloch Sphere Representation

While the Qoppa parameter, Υ_P, was originally defined in the context of the Poincaré sphere to track polarization state evolution, a natural generalization arises when considering its analog in the quantum mechanical Bloch sphere representation.

The Bloch sphere describes the state space of a single qubit:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle, \tag{35}$$

with the corresponding Bloch vector given by:

$$\vec{r} = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta), \tag{36}$$

where θ and ϕ define the position on the unit sphere for pure states, and $\|\vec{r}\| < 1$ for mixed states due to decoherence.

In this context, we define a generalized Bloch-sphere Qoppa parameter, $\Upsilon_B(t)$, as:

$$\Upsilon_B(t) = \frac{\Delta\theta(t)}{d|\vec{r}|/dt}, \tag{37}$$

where $\Delta\theta(t)$ is the angular displacement between successive Bloch vectors, and $|\vec{r}|$ reflects the purity of the quantum state.

Physically, Υ_B measures the rate of angular evolution of the qubit state on the Bloch sphere relative to its change in purity or dynamic energy. Low values of Υ_B suggest smooth, coherent evolution—analogous to synchronized behavior—while high values reflect instability, decoherence, or rapid transitions in quantum state space.

This formulation preserves the original spirit of Qoppa as a dynamical stability metric, but now within the formalism of quantum state geometry. Notably, large excursions of Υ_B may indicate onset of decoherence, non-unitary evolution, or entanglement degradation—features critical to quantum communication and control.

Furthermore, this reinterpretation allows potential connections to geometric and topological quantum concepts. In particular, large values of Υ_B may be indicative of rapid changes in geometric phase or underlying curvature on the Bloch manifold, potentially linking Qoppa to the quantum geometric tensor or Berry curvature.

Thus, Qoppa may serve as an operational probe of geometric features of quantum evolution in optical fiber.

D. Physical Setup and Momentum Space Interpretation

Considering our experimental configuration in which entangled photon pairs are generated via spontaneous parametric down-conversion (SPDC), with photons initialized in horizontal (H) and vertical (V) linear polarization states. These photons are coupled into a polarization-maintaining (PM) optical fiber, which guides their propagation through space.

Suppose given that our PM fiber path in the Sagnac loop is nontrivial—e.g., wound into a loop and following a 3D trajectory— light in the PM fiber nevertheless exits the fiber in the same direction as it entered. The input and output polarization states can then be analyzed via high-precision polarimetry. Crucially, in the semiclassical waveguide picture, the PM fiber then constrains the photon's momentum to remain tangent to the fiber, while the polarization vector lies in the plane orthogonal to the momentum.

This setup defines a natural mapping from the fiber's spatial trajectory to a curve on the **momentum sphere**. The polarization vector, constrained to remain perpendicular and parallel transported along this curve, undergoes a geometric phase shift. This is an interferometric equivalent to a Pancharatnam-Berry phase, [21], [20] proportional by a Gauss mapping to the solid angle enclosed by the momentum path on the sphere:

$$\Phi_{\text{Berry}} = \oint \vec{A}\cdot d\vec{k} = \Omega_{\text{solid angle}}, \tag{38}$$

where $\vec{A}$ is the Berry connection, and $d\vec{k}$ is the infinitesimal change in momentum.

As the fiber traces out its path in space, the momentum vector of the light traces out a path on a sphere in momentum space. The path is closed, since initial and final directions of the light coincide, and the polarization state is a vector tangent to the sphere.

Measurements of the resulting phase shift in previous experiments, done by Bhandari and Samuel, found phase shifts to

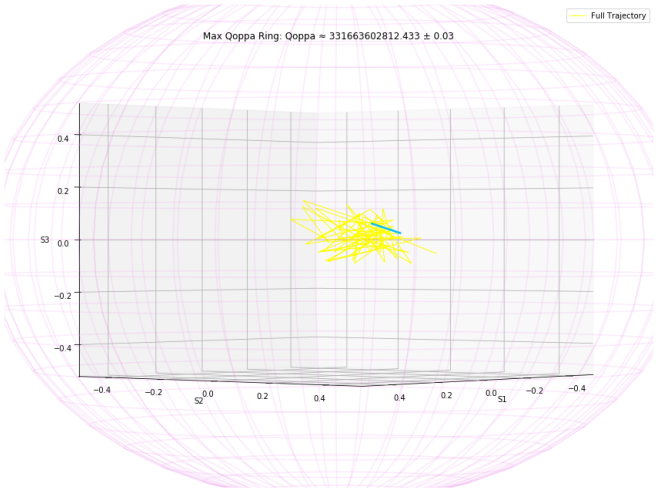


Fig. 16. Polarization evolution under parallel transport on the momentum sphere. The enclosed blue line area corresponds to the Qoppa factor mapped Berry phase accumulated by the polarization trajectory in the fiber loop.

be equal to half the solid angle of the path on the sphere. [19] Their work demonstrated that unitary time evolution was not necessary for the appearance of a topological phase and moreover that the Berry-like phases are a more general phenomenon than envisaged in a purely quantum context.

The Qoppa parameter $\Qoppa_P(t)$, as computed from polarization dynamics, can be viewed as a dynamical quantity sensitive to the geometric evolution of polarization in such a system. In this context, high values of Qoppa correspond to rapid angular displacements in the polarization state, reflecting sharp changes in the underlying Berry curvature:

$$\mathcal{F}_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \tag{39}$$

where $\mathcal{F}_{\mu\nu}$ captures local curvature of the state space.

In experiments, these high-Qoppa regions may form *rings* on the Poincaré sphere or momentum sphere, tracing loci of maximum geometric deviation, indicative of nonlinear optical effects, polarization-mode dispersion, or temporal decoherence.

E. Role of Entanglement and Superposition in the Fiber

When H and V polarization states are entangled across photon pairs and transmitted through the fiber, the combined quantum state evolves in a joint Hilbert space. This superposition persists only as long as phase coherence is maintained across both modes. The fiber acts as a birefringent medium, and even minor deviations in path length or external perturbations can lead to differential phase shifts, which degrade entanglement fidelity.

In the Bloch sphere representation of a single-photon polarization state:

$$|\psi\rangle = \alpha|H\rangle + \beta|V\rangle, \tag{40}$$

the fiber evolution induces trajectory shifts over the sphere, where the coherence of α and β becomes critical. Entanglement thus couples this local trajectory to that of the partner photon.

Decoherence or mode-mixing due to the fiber will result in the Bloch vector contracting (mixed state), and Qoppa, defined as:

$$\Qoppa_B(t) = \frac{\Delta\theta(t)}{d|\vec{r}|/dt}, \tag{41}$$

becomes large in regimes where angular rotation occurs while purity (i.e., length of $\vec{r}$) changes—highlighting instability in the fiber transmission or environmental perturbation.

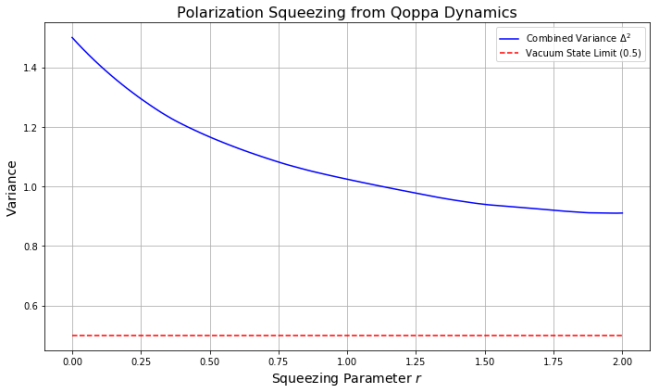


Fig. 17. Plot of $\Qoppa_B(t)$ versus polarization squeezing (length of $\vec{r}$) in PM fiber. Low-purity, high-angle-change regions correspond to rapid decoherence and instability in a quantum fiber channel, towards the vacuum state limit.

X. Conclusions

- Quantum Sagnac Interferometry and tomography is possible using off-the-shelf polarization-maintaining fiber optics components
- The R vs Dop can serve as potential synchronized network benchmark to interpret the state reconstruction wrt polarimetry measures.
- We have demonstrated a diagnostic role of $\Qoppa_P$ as an index of polarization integrity in Fiber Networks that can scale in complexity.

Future research could explore more detailed mapping between the dynamics of the higher-order Kuramoto model for squeezed states in a large fiber network. Investigating the impact of dynamic node coupling strength and synchronization behavior on a network and how Qoppa could be used here would be particularly insightful.

[This material is based upon work supported by Science Foundation Ireland (SFI) and is co-funded under the European Regional Development Fund under Grant Number 13/RC/2077 and 13/RC/2077-P2.]

REFERENCES

[1] Caves, C. M. (1981). Quantum limits on noise in linear amplifiers. *Physical Review D*, 26(8), 1817-1839.

[2] Sharping, J. E., Lee, K. F., Mehaffey, J., Turner, A. C. (2006). Generation of correlated photons in nanoscale silicon waveguides. *Optics Express*, 14(25), 12337-12343.

[3] Tanzilli, S., Simon, C., deMicheli, M. P., D'Ariano, G. M. (2005). On the genesis of polarization-entangled photons. *Physical Review Letters*, 95(6), 063601.

[4] Ou, Z. Y., Mandel, L. (1992). Observation of sub-Poissonian statistical fluctuations in the photon statistics of two-mode coherent light. *Physical Review Letters*, 68(12), 2147-2150.

[5] Xiang, G. Y., Yang, J. (2017). Polarization squeezing and quantum interference in optical fibers. *Quantum Science and Technology*, 2(1), 015003.

[6] Vahlbruch, H., et al. (2016). Detection of squeezed light with low noise photodiodes. *Physical Review Letters*, 117(11), 110501.

[7] Buchmann, L. et al. (2018). Robust polarization squeezing with low-loss fibers. *Optics Express*, 26(22), 29069-29076.

[8] Gottesman, D., Kitaev, A., Preskill, J. (2001). Encoding a qubit in an oscillator. *Physical Review A*, 64(1), 012310.

[9] Heisenberg, W. (1927). Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik. *Zeitschrift für Physik*, 43(3-4), 172-198.

[10] Bravyi, S., König, R. (2005). Quantum codes from GKP states. *Physical Review A*, 71(3), 032316.

[11] Niset, J., Fiurasek, J. (2009). Sensing with squeezed light. *Physical Review A*, 80(4), 045803.

[12] Lupo, C., Pirandola, S. (2018). Phase-insensitive amplification of quantum states. *Physical Review Letters*, 120(13), 130503.

[13] On Using The Path Integral Formalism to Interpret Synchronization in Quantum Graph Networks JTM Campbell, Department of Electronic Engineering, Maynooth University, Ireland.

[14] Wang, X. et al. (2019). Experimental demonstration of quantum state transfer via polarization squeezing in optical fibers. *Nature Communications*, 10, 78.

[15] Caruso, F., et al. (2011). Quantum Information Transfer through a Coherent Beam. *Physical Review Letters*, 106(9), 090501.

[16] O. Bedroya *et al.*, "Resource-Efficient Real-Time Polarization Compensation for MDI-QKD with Rejected Data," *arXiv preprint*, arXiv:2209.02707, Sept. 2022.

[17] Y. Zou, H. Lai, and W. Zhang, "A New Recurrence-Network-Based Time Series Analysis Approach for Characterizing System Dynamics," *Entropy*, vol. 21, no. 1, p. 45, Jan. 2019.

[18] M. V. Berry, "Quantal phase factors accompanying adiabatic changes," *Proc. R. Soc. Lond. A*, vol. 392, no. 1802, pp. 45–57, 1984.

[19] R. Bhandari and J. Samuel, "Observation of topological phase by use of a laser interferometer," *Phys. Rev. Lett.*, vol. 60, no. 13, pp. 1211–1213, 1988.

[20] R. Y. Chiao and Y. S. Wu, "Manifestations of Berry's topological phase for the photon," *Phys. Rev. Lett.*, vol. 57, no. 8, pp. 933–936, 1986.

[21] S. Pancharatnam, "Generalized theory of interference and its applications," *Proc. Indian Acad. Sci. A*, vol. 44, pp. 247–262, 1956.

APPENDIX

This appendix provides a detailed overview of key figures of merit implemented for analyzing and simulating quantum-enhanced stabilization of fiber networks utilizing Gottesman-Kitaev-Preskill (GKP) squeezed qubits. The simulation code, (available at https://github.com/MuonRay/QuantumNetworkSimulations/blob/main/FOM leverages quantum optics tools (via `qutip`) and numerical libraries (`numpy` and `matplotlib`) to visualize polarization squeezing, fibered interferometry phase shifts, quantum state discrimination, and sensitivity measurements hich compare favorably with planned figure of merit tests for the experimental test setup.

This simulation is based upon the numerical analysis of a fibered networked Lagrangian in order to determine polarization shifts wrt fiber network synchronization, to determine the Qoppa factor.

A. Polarization Shifts as a Doppler-like Effect in the Network Lagrangian Framework

Building upon the established relationship between frequency shifts and the time derivative of the action [13], we propose a novel interpretation of polarization dynamics in optical fiber networks using a network Lagrangian formalism. Specifically, we model the polarization shift ΔP as being analogous to a Doppler-like response to variations in the action S of a distributed physical system.

In relativistic mechanics and wave propagation in media, the Doppler effect arises due to changes in observed frequency resulting from relative motion or time-varying media. Analogously, in complex photonic systems—such as fiber optic networks—external physical influences (e.g., temperature gradients, mechanical stress, or structural imperfections) induce shifts not only in frequency but also in polarization states.

We postulate that such changes in the polarization of transmitted light can be captured through a Lagrangian-based framework by introducing a time-dependent action $S(t)$, whose rate of change governs the observable polarization dynamics. That is, we write:

$$\boxed{\Delta P \propto \mathfrak{Q}_P \frac{\partial S}{\partial t}} \tag{42}$$

where:

- ΔP is the observed shift in polarization state,
- $\mathfrak{Q}_P$ is a phenomenological proportionality constant that encapsulates the sensitivity of the system to physical perturbations,
- $\frac{\partial S}{\partial t}$ is the time derivative of the action functional S over the networked system.

a) Interpretation:: This equation suggests that any temporal variations in the system's action—arising from environmental or structural changes—will lead to measurable shifts in polarization. The action S may be constructed as an integral over a network Lagrangian $\mathcal{L}_{\text{net}}$, defined over the network graph $G = (V, E)$, as:

$$S = \int_{t_0}^{t_1} \sum_{e \in E} \mathcal{L}^e_{\text{net}}(q_e, \dot{q}_e, t)\, dt \tag{43}$$

Here, q_e are generalized coordinates associated with each edge e (e.g., field amplitudes, phase, polarization state, etc.), and $\mathcal{L}^e_{\text{net}}$ is the local Lagrangian density per edge.

b) Doppler-Like Analogy:: In traditional Doppler theory, a frequency shift Δf is given by:

$$\Delta f \propto \frac{d\phi}{dt}$$

where ϕ is the phase. By analogy, here, we interpret ΔP as a shift resulting from temporal modulations of the entire system's action, which can be seen as an effective "phase" for the polarization field in a networked context.

c) Physical Relevance:: This relation becomes particularly useful in real-world fiber networks where birefringence, mode coupling, and environmental perturbations nontrivially affect signal integrity. By tracking $\partial S/\partial t$, one could potentially infer or control polarization drift in sensitive systems, such as quantum key distribution (QKD), coherent communications, or fiber-based sensors.

B. Electromagnetic Lagrangian Modeling and Numerical Simulation of Polarization Shift

To ground our polarization shift model in physical terms, we extend the network action formalism to electromagnetic (EM) field dynamics within optical fibers. The Lagrangian density for an EM field in a dielectric medium is given by:

$$\mathcal{L}_{\text{fiber}} = \frac{1}{2}\left(\varepsilon_r \varepsilon_0 |\mathbf{E}|^2 - \frac{1}{\mu_r \mu_0}|\mathbf{B}|^2\right) \tag{44}$$

where $\mathbf{E}$ and $\mathbf{B}$ are the local electric and magnetic fields within the fiber, and ε_r, μ_r are the relative permittivity and permeability of the fiber material, which may be functions of space or time due to environmental perturbations.

For a network of fibers modeled as a graph $G = (V, E)$, the total action S becomes:

$$S = \sum_{e \in E} \int_{t_0}^{t_1} \int_{z_e} \mathcal{L}^e_{\text{fiber}}(z, t)\, dz\, dt \tag{45}$$

We hypothesize that changes in polarization ΔP result from the time derivative of the action, i.e.,

$$\boxed{\Delta P(t) \propto \Omega_P \frac{\partial}{\partial t}\left[\sum_{e \in E} \int_{z_e} \mathcal{L}^e_{\text{fiber}}(z, t)\, dz\right]} \tag{46}$$

a) Numerical Simulation Framework:: To simulate this, we discretize each fiber edge e into spatial segments of length Δz, and discretize time with steps Δt. The local Lagrangian per segment i on edge e is:

$$\mathcal{L}^e_i(t) = \frac{1}{2}\left(\varepsilon^e_i |\mathbf{E}^e_i(t)|^2 - \frac{1}{\mu^e_i}|\mathbf{B}^e_i(t)|^2\right) \tag{47}$$

Then the instantaneous polarization shift is approximated as:

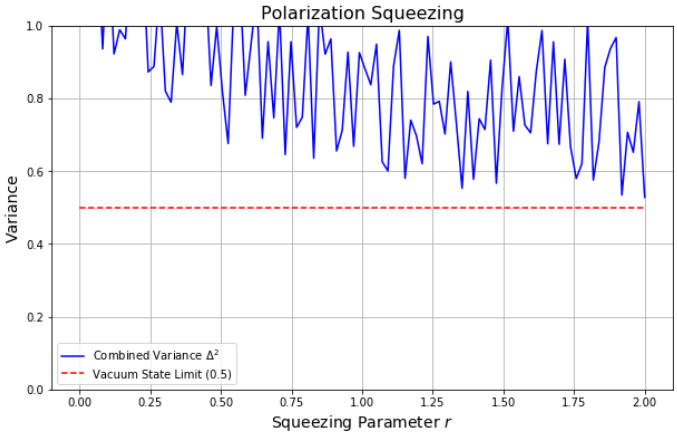
$$\Delta P(t) \approx \Omega_P \sum_{e \in E} \sum_i \mathcal{L}^e_i(t) \cdot \Delta z \tag{48}$$

Field values $\mathbf{E}^e_i(t)$ and $\mathbf{B}^e_i(t)$ can be updated using a finite-difference time-domain (FDTD) scheme or beam propagation method (BPM), depending on desired accuracy and computational complexity.

b) Interpretation:: This model enables tracking polarization drift as a function of time-varying field energy in the network. As environmental conditions evolve—altering local ε, μ, or inducing modal dispersion—these changes are captured in $\mathcal{L}_{\text{fiber}}$, and consequently reflected in $\Delta P(t)$.

C. non phase tuned Polarization Squeezing

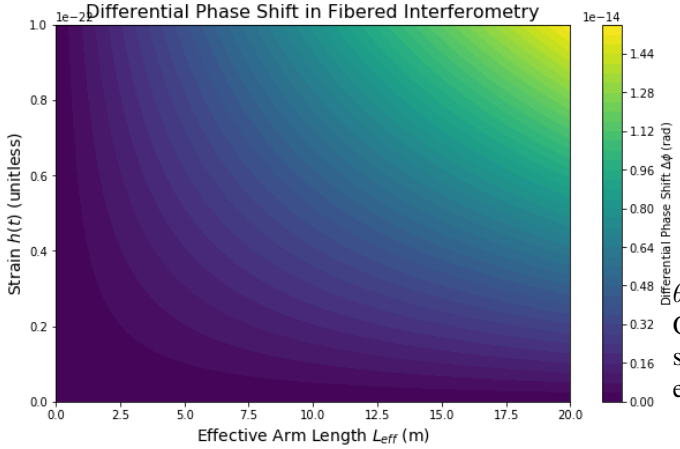
This section provides an untuned polarization squeezing—a quantum resource used to reduce quantum noise in specific polarization quadratures, which can enhance the stability of fiber communication channels.



Explanation: - `r_values`: Range of squeezing parameters r, which quantify the degree of squeezing. - `Delta_x_squared`: Variance in the squeezed quadrature, exponentially decreasing with r, representing reduced quantum noise. - `Delta_y_squared`: Variance in the orthogonal quadrature, modeled with some randomness to simulate measurement uncertainty. - The plot displays how the combined variance decreases below the vacuum shot-noise limit (0.5) at higher squeezing levels, illustrating potential, and hence motivation, for noise reduction in polarization states used for fiber transfer of qubits, wherin stabilization by means of EOM tuned interferometry comes in.

D. Fibered Interferometry Phase Shift

This segment simulates the differential phase shift $\Delta\phi$ induced by gravitational wave strains in a fibered interferometric setup, critical for detecting minute disturbances affecting fiber stability.



Plot Explanation: - `k`: Wavenumber corresponding to the laser wavelength, setting the phase sensitivity scale. - `L_eff` and `h_t`: Mesh grids representing the variable arm length and gravitational wave strain, respectively. - `Delta_phi`: Calculated phase shift, which depends linearly on both parameters, illustrating how minute strains influence phase measurements in the fiber network.

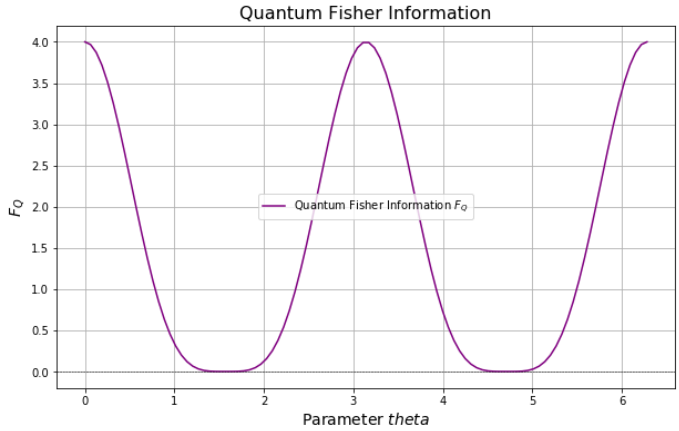
E. Quantum State Discrimination and Quantum Fisher Information

Quantum state discrimination techniques enable the detection of small differences in quantum states, essential for fiber stabilization at the quantum limit.

```
def plot_quantum_fisher_information():
    theta_values = np.linspace(0, 2 * np.pi, 1
    psi = np.sin(theta_values)
    d_psi_d_theta = np.cos(theta_values)

    F_Q = 4 * (np.abs(d_psi_d_theta)**2 - np.a

    plt.figure(figsize=(10, 6))
    plt.plot(theta_values, F_Q, label='Quantum
    plt.xlabel('Parameter $theta$', fontsize=1
    plt.ylabel('$F_Q$', fontsize=14)
    plt.title('Quantum Fisher Information', fc
    plt.grid()
    plt.axhline(0, color='black', linewidth=0.
    plt.legend()
    plt.show()
```

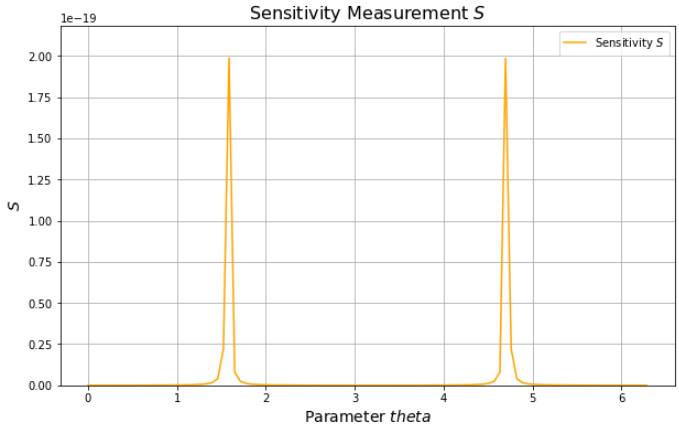


Explanation: - `psi`: The quantum state parametrized by θ, modeled here as a sine function for simplicity. - `F_Q`: The Quantum Fisher Information quantifies the sensitivity of the state to changes in θ, indicating the potential precision in estimating θ.

—

F. Sensitivity Measurement

This section computes the sensitivity S of the measurement process, which is crucial for detecting weak signals such as vibrational waves in fiber networks.



G. Simulator Required Detector Speed for Phase Shift Resolution beyond Heisenberg Limit